SmartAg Ops Advisor — Product Requirements Document v4.2

Oct 25, 2026 · @Boopathi

SmartAg Ops Advisor is one guest flow: a small tomato grower enters a batch and its costs and sees, per 14 kg crate, which comparable market leaves the most money and whether the batch covers picking and selling costs; when the data cannot support a ranking, it says why.

Product principle: show what the source supports, label where and when each price came from, and never present an estimate as a guaranteed price.

Version 4.2 (consolidates v4.0 and v4.1) · traces to MRD v3.2 · React + Vite, FastAPI, Neon PostgreSQL, no ML · data: labelled sample or dated snapshot · demo 25 Oct 2026, 10:00.

Scope decision: the existing Master Application (login, crop stock, trucks, bookings, tracking, reports, AI chat) stays in the code but is hidden in demo mode; only the comparison flow is demoed, and its fixed cost defaults and 35 km/h ETA are not used.

Won't build in this release: login in the demo path, Tamil UI, chatbot, price forecasts or any ML, alerts, GPS, truck booking and tracking, route ETA, payments, potato.

1. Problem and solution

Small tomato growers commit money to picking and transport before they know what a market will leave them; SmartAg turns the prices and costs they already know into a per-crate answer before that money is spent.

Problem: a grower near Oddanchatram cannot quickly tell which reachable market leaves the most after picking, transport and selling costs, or whether the batch covers them at all; prices arrive undated and in mixed units, and in Dindigul in March 2026 Rs 80 a box to pick left Rs 20–70 from a Rs 100–150 box (evidence in MRD v3.2 §1).

Solution: dated, labelled price cards; conversion to Rs per kg and per 14 kg crate; an estimate after the trip, commission and handling the farmer entered; the Harvest Cover Check per crate; and a ranking that names a winner only when every check passes, with a reminder that a reported price is a benchmark, not a buyer's offer.

2. Features

The prototype has three Must-Have features; every functional requirement sits under one of them, with FR numbers kept from v4.0 and FR-21 new.

Must-Have	FR	Acceptance
1. Trustworthy price view	01–02	Guest access; badge always on; only configured crop–market pairs
	03	Card: market, source, date, modal (min–max), unit, type; missing = "not reported"
	04–05	Older than 2 days (IST) not ranked; kg, quintal, tonne, 14 kg crate convert; else "unit unclear"
	06–07	Wholesale and Uzhavar Sandhai never mixed; same crop, measure and date to rank
	15–16	Sample never shown as reported; source error ≠ empty result
2. Estimate after costs	08–09	Quantity > 0; blank cost is unknown, zero needs "no cost"; trip cost per market, trips and load fill
	10–11	Optional commission, loading, handling; "estimate after entered costs"; round on display only
	21	**Harvest Cover Check:** left per crate and cover price per market; "no market covers costs" message; never advises harvesting; ranking unchanged
3. Honest ranking	12–14	One state in §4 order; winner only when RANKED, with lead, dates and exclusions; assumptions strip

Should: FR-17 strings externalised (Tamil after native review), FR-18 copy result as text. **Could:** FR-22 break-even line, FR-23 calculation log. **Won't:** FR-19 profiles and alerts, FR-20 chatbot, and the Won't list above.

3. Calculation rules

All calculations are pure, deterministic functions with no hidden defaults: every input is farmer-entered or read from a dated record.

Eligibility: a price ranks only if the source run succeeded; source, date, unit and type are known; it is 2 days old or less; the unit converts; min ⩽ modal ⩽ max; and crop, measure, type and date match the other markets. Each failure shows a plain reason.

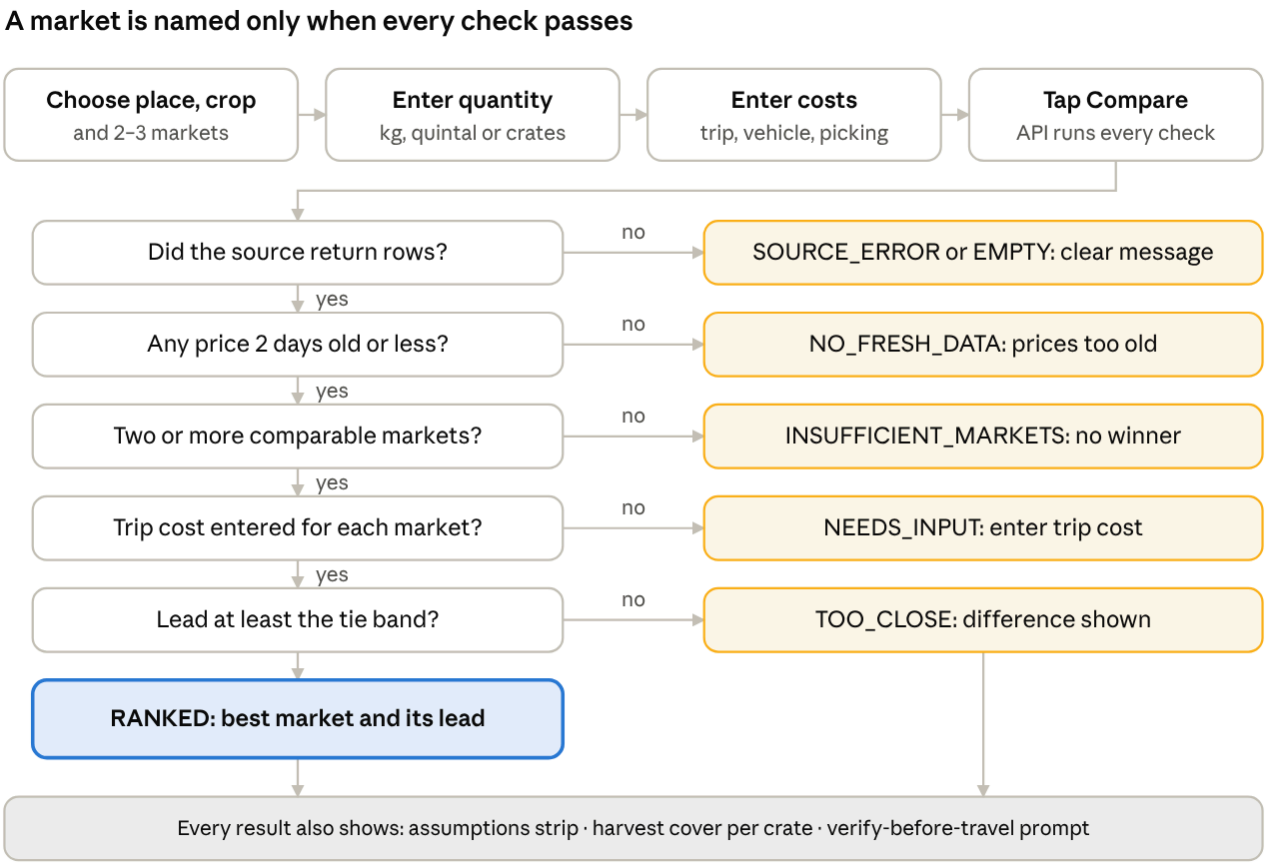
Measure	Formula
Price per kg; gross	modal ÷ kg per unit; × quantity in kg
Trips; transport; load fill	ceil(quantity ÷ capacity); trips × trip cost; quantity ÷ (trips × capacity)
Other costs; estimate	loading + handling + commission (% × gross, or flat); gross − transport − other
Crates; harvest cost	quantity ÷ 14; crates × (picking + packing per crate)
Left per crate	(estimate − harvest cost) ÷ crates
Cover price per crate	(transport + flat other + harvest cost) ÷ (crates × (1 − commission %))

Worked example: 700 kg (50 crates), picking Rs 80 per crate, one trip each. Nearby at Rs 120 per crate with a Rs 1,800 trip leaves Rs 4 per crate (cover price Rs 116); farther at Rs 150 with a Rs 3,000 trip leaves Rs 10 (cover price Rs 140), and ranks first by Rs 300.

Ranking: sort eligible estimates highest first. TOO_CLOSE when the top two differ by less than max(Rs 100, 2% of the higher); equal to the band is ranked.

4. Workflow and result states

The farmer completes four steps on one page; the API then runs five checks in order, and the first that fails decides the state.

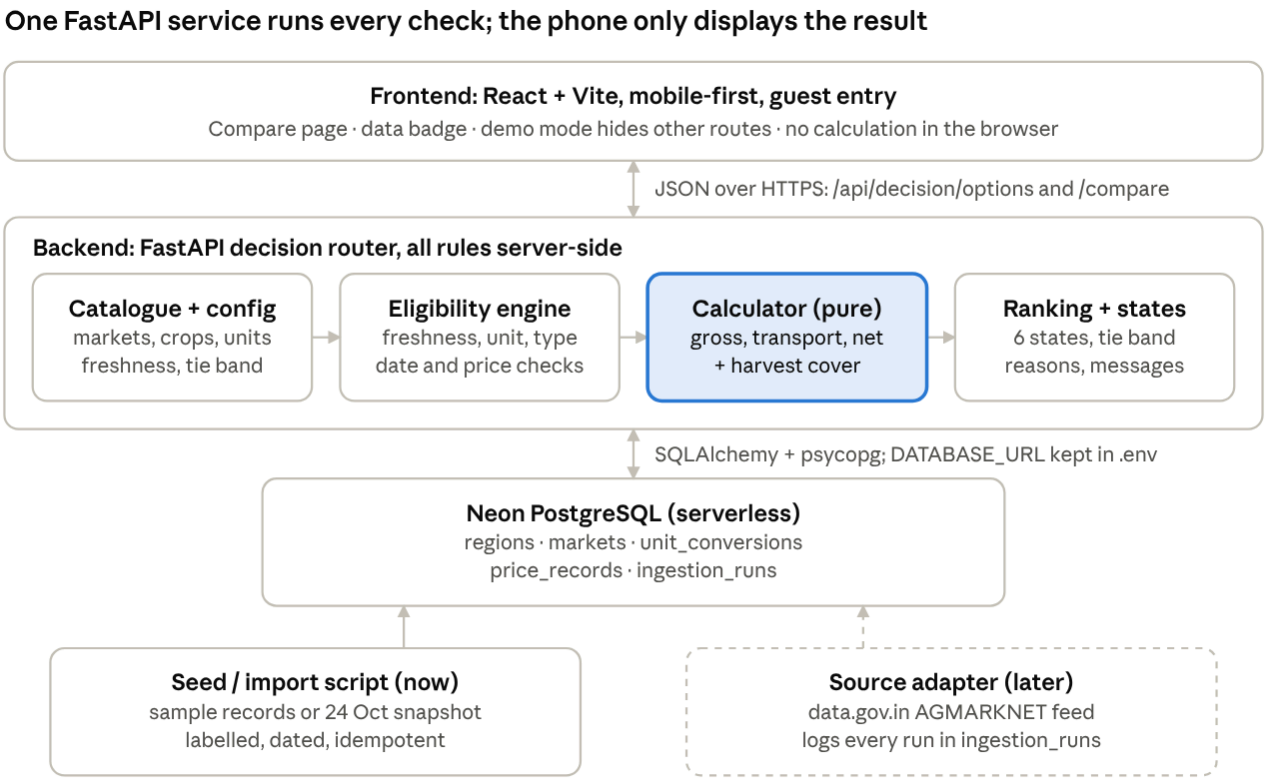


Workflow · 4 steps, 5 checks, 6 states

Messages: SOURCE_ERROR "Could not retrieve data. Try again later." · EMPTY "No matching records found." · NO_FRESH_DATA "Prices are too old to compare." · INSUFFICIENT_MARKETS "Ranking unavailable: fewer than two fresh comparable markets." · NEEDS_INPUT "Enter trip cost to see net." · TOO_CLOSE "These two are too close to call (difference Rs N)." · RANKED "Estimated best: Market X, about Rs N more than Market Y after the costs you entered."

5. Architecture

Three layers inside the existing agri-project repo: a thin React page, a new FastAPI decision router that holds every rule, and Neon PostgreSQL loaded by a seed script; the old comparison code is left untouched because bookings use it.



Architecture · 3 layers, 4 backend modules

API: `POST /api/decision/compare` takes the inputs and returns the state, message, winner (null unless RANKED), lead, tie band, one card per market and the costs not included; `POST /api/auth/guest` issues a demo token.

Compare page, top to bottom: data badge ("Sample data as of 25 Oct 2026" or "Reported 24 Oct 2026, AGMARKNET"); inputs (place, crop, 2–3 markets, quantity and unit, vehicle capacity, trip cost or "no cost" per market, optional commission, loading, handling, picking and packing per crate); decision box; Harvest Cover panel; market cards (modal and min–max, unit, date, source, freshness in words, Rs/kg and Rs/crate, gross, transport, estimate, load fill); excluded markets with reasons; assumptions strip. Rupees use Indian grouping and no status relies on colour alone.

6. Data

Every place-specific fact lives in rows, not code, so a new district or crop is added by inserting rows and enabling the pair once it has two fresh, comparable markets.

Tables: `regions`; `markets` (type: wholesale, uzhavar_sandhai, unknown; enabled); `unit_conversions` (kg 1, quintal 100, tonne 1,000, crate 14); `price_records` (market, crop, variety, grade, report_date, min, max, modal, unit, source, is_sample, run_id, raw_ref; unique on market + crop + date + variety + grade + source); `ingestion_runs` (SUCCESS, EMPTY or ERROR).

Farmer inputs are not stored; config holds freshness 2 days, tie band Rs 100 or 2%, Asia/Kolkata, `demo_as_of` 2026-10-25 and `data_mode`. **Data mode:** the automated pull returned no valid rows and the historical subset has no Dindigul market, so the 173-row 24 Oct snapshot is used only if its unit and Dindigul coverage are confirmed; otherwise the sample seed below.

Sample seed (8 records, names and types to be confirmed by the BA): Oddanchatram Rs 120/crate, Dindigul Rs 1,050/quintal and Palani Rs 150/crate, all wholesale on 24 Oct (rankable); Dindigul Uzhavar Sandhai Rs 13/kg (not mixed with wholesale); Vedasandur 23 Oct (different date); Batlagundu 21 Oct (stale); Nilakottai per box (unit unclear); Natham modal 130 above max 120 (price invalid).

7. Quality requirements and success metrics

Non-functional: works at phone width without horizontal scroll; result in under 3 s; hand-checked cases within Rs 1 with no rounding before ranking; timeout, empty and malformed data give distinct states; no personal data stored and secrets kept server-side; labelled inputs and status in words; thresholds, conversions and market types in config behind a source adapter.

Success metrics: 18 of 18 acceptance tests pass before the demo; all 6 states shown live or from seed data; zero winners on invalid data; first result in under 60 s with costs known. After the hackathon, in a 5-farmer pilot: 4 of 5 explain the ranking and name one cost not included, and 3 of 5 would check the per-crate answer before paying for picking.

8. Acceptance tests

All 18 run as automated tests; a second team member rechecks AT-05, AT-14, AT-17 and AT-18 by hand.

ID	Input	Expected
AT-01	A Rs 2,800/quintal; B Rs 350/crate; 500 kg	Rs 28 and 25/kg; gross 14,000 and 12,500
AT-02	Rs 350/box, weight unknown	"Unit unclear"
AT-03	A 1 day old, B 3 days	B stale; INSUFFICIENT_MARKETS
AT-04	Wholesale vs Uzhavar Sandhai	Not ranked together

ID	Input	Expected
AT-05	500 kg at Rs 25/kg; trip 1,800	Estimate 10,700
AT-06	AT-05 + 10% commission	1,250; estimate 9,450
AT-07	Trip 1,800; capacity 1,000; 600 kg	1 trip; Rs 3/kg; fill 60%
AT-08	Same, 1,500 kg	2 trips; 3,600; Rs 2.40/kg; 75%
AT-09	One trip cost blank	NEEDS_INPUT
AT-10	Timeout vs zero rows	Two different messages
AT-11	One eligible market	INSUFFICIENT_MARKETS
AT-12	13,200 vs 13,000	TOO_CLOSE (band 264)
AT-13	13,200 vs 12,936	RANKED (equals band)
AT-14	500 kg; A 28/kg trip 2,500; B 25, 1,800; C 26, 1,200	C first by 300 over A
AT-15	Sample data	Badge on every screen
AT-16	Quantity ⩽ 0 or negative cost	Field error, no result
AT-17	50 crates; picking 80; 120 trip 1,800 vs 150 trip 3,000	Left 4 and 10; cover 116 and 140; far first by 300
AT-18	AT-17 at 110 and 135 per crate	Left −6 and −5; "no market covers"; TOO_CLOSE

9. Traceability, risks and delivery

Every Must requirement traces from an MRD need to a feature and a test; the Story column is filled from the backlog.

MRD need	Must	FR	Tests	Story
MR-01 crop, MR-02 provenance, MR-03 units	1	FR-02, 03, 05	AT-01, 02	
MR-04 comparability, MR-05 freshness	1	FR-04, 06, 07	AT-03, 04	
MR-10 failures, MR-11 demo label	1	FR-01, 15, 16	AT-10, 15	
MR-07 quantity and costs	2	FR-08, 09, 10	AT-07, 08, 09, 16	
MR-08 estimate and rank	2, 3	FR-11, 12	AT-05, 06, 12, 13, 14	
MR-15 harvest cover	2	FR-21	AT-17, 18	
MR-06 insufficient data, MR-09 explain	3	FR-12, 13, 14	AT-03, 11, 14	
MR-12 Should; MR-13, MR-14 Later	—	FR-17; Won't	—	—

Risks and controls: hidden Master Application features → demo mode hides routes (Dev); sample mistaken for live → badge on every screen (UI/UX); over-trust in a ranking → TOO_CLOSE, exclusions and assumptions strip (BA); arithmetic error → tests before each rehearsal (QA); Neon idle or network down → open early, dated snapshot fallback (Dev).

Delivery sequence: S0 audit and data-mode decision → S1 decision engine and tests → S2 API and guest token → S3 Compare page, data badge and demo mode → S4 README, test checklist and backup recording → rehearsal.

Definition of done: a stranger runs it from the README in under 5 minutes; every price shows source, date, unit and type; all 18 tests pass; every insufficient-data case shows an honest state instead of a winner; no hidden cost default or ETA reaches the result.